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Google Scholar Link: <https://scholar.google.com/citations?user=wNEYBrAAAAAJ&hl=en>

BIO AND RESEARCH SUMMARY

My general research interests are in the areas of artificial intelligence (AI) and machine learning (ML) where I focus on advancing foundations of AI/ML to solve challenging real-world problems. The overarching theme of my research program is AI to Accelerate Scientific Discovery and Engineering Design. Specific topics include:

- Sequential decision-making under uncertainty in a data-efficient manner (Bayesian optimization and reinforcement learning).
- Probabilistic modeling over combinatorial structured data in small supervised data settings.
- Applications: accelerate the discovery of nanoporous materials for sustainability applications (e.g., carbon capture and storage/separation of gases); accelerate the design of effective, safe, and low-cost drugs/vaccines; and design of high-performance and low-power hardware.

PROFESSIONAL EXPERIENCE

Assistant Professor (Tenure Track)

Department of Computer Science and Engineering, University of Minnesota

August 2024–

Minneapolis, Minnesota

EDUCATION

Ph.D., Computer Science

Washington State University

Advisor: *Prof. Jana Doppa*

Thesis: Adaptive Experimental Design for Optimizing Combinatorial Structures

2023 Voiland College of Engineering and Architecture, Outstanding PhD Dissertation Award

2018–2024

Pullman, Washington

MS, Computer Science

Washington State University

Thesis: A Learning to Search Framework for Multi-Objective Optimization of Manycore Systems Design

2020 WSU Outstanding Dissertation Award. (Nominated for WAGS Outstanding Innovation in Technology Award)

2018–2020

Pullman, Washington

Bachelor of Technology, Mathematics and Computing

Delhi Technological University

2013–2017

New Delhi, India

AWARDS AND HONORS

- NSF CAREER Award, 2026
- Selected for [IJCAI Early Career Spotlight](#), 2026
- Deployed Application Award, Innovative Applications of Artificial Intelligence (IAAI), 2026
- Top Reviewer Award, Neural Information Processing Systems (NeurIPS), 2025
- AAAI New Faculty Highlights Speaker, 2025
- [Voiland College of Engineering, Outstanding PhD Dissertation Award](#), 2023
- Rising Stars in AI, KAUST AI Initiative, 2023
- Selected for [Heidelberg Laureate Forum](#), 2022
- [Voliland College of Engineering, Outstanding Graduate Research Assistant Award](#), 2022
- Top Reviewer Award, International Conference on Machine Learning (ICML), 2021
- Outstanding Reviewer Award, International Conference on Learning Representations (ICLR), 2021
- Top Reviewer Award, International Conference on Machine Learning (ICML), 2020
- WSU Outstanding Dissertation Award (Nominee for WAGS Outstanding Innovation in Tech. Award), 2020

- WSU GPSA Graduate Research Assistant Excellence Award, 2020
- Voiland College of Engineering, Outstanding Teaching Assistant in Computer Science Award, 2020
- Mahmoud Dillsi Graduate Fellowship, 2020
- Alfred Suksdorf Fellowship, Washington State University, 2018-2020

PUBLICATIONS

Conference Papers

23. [AAAI'26] Nanoporous Materials Discovery via Search Bias-Guided Surrogate Modeling. Azza Fadhel, Yassine Chemingui, Minh Hoang, Aryan Deshwal, Trong Nghia Hoang, Jana Doppa. Fortieth AAAI Conference on Artificial Intelligence, AI for Social Impact Track (AAAI), 2026.
22. [AAAI/IAAI'26] Azza Fadhel, Nathaniel W. Zuckschwerdt, Aryan Deshwal, Susmita Bose, Amit Bandyopadhyay, Jana Doppa. Discovery of Feasible 3D Printing Configurations for Metal Alloys via AI-Driven Adaptive Experimental Design. Fortieth AAAI Conference on Artificial Intelligence and Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence (AAAI/IAAI). *Innovative Deployed Application Award*, 2026.
21. [NeurIPS'25] Yassine Chemingui, Aryan Deshwal, Alan Fern, Thanh Nguyen-Tang, Jana Doppa. Online Optimization for Offline Safe Reinforcement Learning. Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.
20. [NeurIPS'25] Seunghee Ryu, Donghoon Kwon, Seongjin Choi, Aryan Deshwal, Seungmo Kang, Carolina Osorio. BO4Mob - Bayesian Optimization Benchmarks for High-Dimensional Urban Mobility Problem. Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track (NeurIPS), 2025.
19. [EMNLP'25] Gaoxiang Luo, Aryan Deshwal. Bayesian Exemplar Search for Efficiently Exploring the Accuracy-Calibration Pareto Frontier. Thirtieth Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.
18. [AAAI'25] Yassine Chemingui, Aryan Deshwal, Honghao Wei, Alan Fern, Janardhan Rao Doppa. Constraint-Adaptive Policy Switching for Offline Safe Reinforcement Learning. *Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
17. [ICML'24] Minh Hoang, Azza Fadhel, Aryan Deshwal, Janardhan Rao Doppa, Nghia Hoang. Learning Surrogates for Offline Black-Box Optimization via Gradient Matching. *Forty-first International Conference on Machine Learning (ICML)*, 2024.
16. [IJCAI'24] Mohammed Amine Gharsallaoui, Bhupinderjeet Singh, Aryan Deshwal, Yan Yan, Kirti Rajagopalan, Ananth Kalyanaraman, Janardhan Rao Doppa. Streamflow Prediction with Uncertainty Quantification for Water Management: A Neuro-Symbolic Learning Approach. *Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.
15. [AAAI'24] Yassine Chemigui, Aryan Deshwal, Nghia Hoang, Janardhan Rao Doppa. Offline Model-based Black-Box Optimization via Policy-guided Gradient Search. *Thirty-Eighth AAAI Conference on Artificial Intelligence (AAAI)*, 2024.
14. [AISTATS'23] Aryan Deshwal, Sebastian Ament, Maximilian Balandat, Eytan Bakshy, Janardhan Rao Doppa, and David Eriksson. Bayesian Optimization over High-Dimensional Combinatorial Spaces via Dictionary-based Embeddings. *Twenty-sixth International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023.
13. [AAAI'22] Aryan Deshwal, Syrine Belakaria, Janardhan Rao Doppa, Dae Hyun Kim. Bayesian Optimization over Permutation Spaces. *Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI)*, 2022.
12. [NeurIPS'21] Aryan Deshwal, Janardhan Rao Doppa. Combining Latent Space and Structured Kernels for Bayesian Optimization over Combinatorial Spaces. *Thirty-Fifth Conference on Neural Information Processing Systems (NeurIPS)*, 2021.
11. [ICML'21] Aryan Deshwal, Syrine Belakaria, Janardhan Rao Doppa. Bayesian Optimization over Hybrid Spaces. *Proceedings of Thirty-eighth International Conference on Machine Learning (ICML)*, 2021.

10. [DAC'21] Aryan Deshwal, Syrine Belakaria, Ganapati Bhat, Janardhan Rao Doppa, and Partha Pratim Pande. Learning Pareto-Frontier Resource Management Policies for Heterogeneous SoCs: An Information-Theoretic Approach. *Proceedings of 58th ACM/IEEE Design Automation Conference (DAC)*, 2021.
9. [AAAI'21] Aryan Deshwal, Syrine Belakaria, Janardhan Rao Doppa. Mercer Features for Efficient Combinatorial Bayesian Optimization. *Proceedings of Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI)*, 2021.
8. [AAAI'20] Aryan Deshwal, Syrine Belakaria, Janardhan Rao Doppa, and Alan Fern. Optimizing Discrete Spaces via Expensive Evaluations: A Learning to Search Framework. *Proceedings of Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI)*, 2020.
7. [AAAI'20] Syrine Belakaria, Aryan Deshwal, Nitthilan Kannappan Jayakodi, and Janardhan Rao Doppa. Uncertainty-Aware Search Framework for Multi-Objective Bayesian Optimization. *Proceedings of Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI)*, 2020.
6. [AAAI'20] Syrine Belakaria, Aryan Deshwal, and Janardhan Rao Doppa. Multi-Fidelity Multi-Objective Bayesian Optimization: An Output Space Entropy Search Approach. *Proceedings of Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI)*, 2020.
5. [DATE'20] Syrine Belakaria[†], Zhiyuan Zhou[†], Aryan Deshwal, Janardhan Rao Doppa, Deuk Heo, and Partha Pratim Pande. Design of Multi-Output Switched-Capacitor Voltage Regulator via Machine Learning. *Proceedings of Twenty-Third ACM/IEEE Design, Automation and Test in Europe Conference (DATE)*, 2020.
[†] denotes equal contribution.
4. [CODES'19] Paul Bogdan, Fan Chen, Aryan Deshwal, Janardhan Rao Doppa, Biresh Kumar Joardar, Hai (Helen) Li, Shahin Nazarian, Linghao Song, Yao Xiao. Taming Extreme Heterogeneity via Machine Learning based Design of Autonomous Manycore Systems. *International Conference on Hardware/Software Codesign and System Synthesis Companion (CODES)*, pp 21:1-21:10, 2019.
3. [NeurIPS'19] Syrine Belakaria, Aryan Deshwal, and Janardhan Rao Doppa. Max-value Entropy Search for Multi-Objective Bayesian Optimization. *Proceedings of Thirty-Third International Conference on Neural Information Processing Systems (NeurIPS)*, pp 7823-7833, 2019.
2. [IJCAI'19] Aryan Deshwal, Janardhan Rao Doppa, and Dan Roth. Learning and Inference for Structured Prediction: A Unifying Perspective. *Proceedings of Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI)*, pp 6291-6299, 2019.
1. [IJCAI'19] Chao Ma, F A Rezaur Rahman Chowdhury, Aryan Deshwal, Md Rakibul Islam, Janardhan Rao Doppa, and Dan Roth. Randomized Greedy Search for Structured Prediction: Amortized Inference and Learning. *Proceedings of Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI)*, pp 5130-5138, 2019.

Journal Papers

10. [Langmuir'26] Makaila Hammond, Qia Ke, Aryan Deshwal, Cory Simon, Alison Bain. Adaptive, Bayesian Experimental Design to Efficiently Determine the Critical Micelle Concentration of a Surfactant. *Langmuir*, 2026.
9. [NPJ'25] AI-enabled Scientific Revolution in the Age of Generative AI – Second NSF Workshop Report. Anuj Karpatne, Aryan Deshwal, Xiaowei Jia, Wei Ding, Michael Steinbach, Aidong Zhang, Vipin Kumar. *npj Artificial Intelligence*, 2025.
8. [Advanced Materials'24] Eric Chen, Alaleh Ahmadian, Sonja Sparks, Aryan Deshwal, Janardhan Rao Doppa, Kaiyan Qiu. Machine Learning Enabled Design and Optimization for 3D-Printing of High-Fidelity Presurgical Organ Models. *Advanced Materials Journal*, 2024.
7. [Digital Discovery'23] Nickolas Gantzler, Aryan Deshwal, Janardhan Rao Doppa, Cory Simon. Multi-fidelity Bayesian Optimization of Covalent Organic Frameworks for Xenon/Krypton Separations. To appear in *Digital Discovery Journal*, *Royal Society of Chemistry*, 2023.
6. [TODAES'23] Gaurav Narang, Aryan Deshwal, Janardhan Rao Doppa, Raid Ayoub, Mike Kishnivesky, Partha Pratim Pande. Dynamic Power Management in Large Manycore Systems: A Learning-to-Search Framework. *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 28(5): 84:1-84:21, 2023.

5. [TCAD'22] Biresh Kumar Joardar, Aryan Deshwal, Janardhan Rao Doppa, Partha Pratim Pande, Krishnendu Chakrabarty. High-Throughput Training of Deep CNNs on ReRAM-based Heterogeneous Architectures via Optimized Normalization Layers. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)* 41(5): 1537-1549, 2022.
4. [MSDE'21] Aryan Deshwal, Cory Simon, Janardhan Rao Doppa. Bayesian Optimization of Nanoporous Materials. *Molecular Systems Design and Engineering Journal, Royal Society of Chemistry*, 6, 1066-1086, 2021.
3. [JAIR'21] Syrine Belakaria, Aryan Deshwal, Janardhan Rao Doppa. Output Space Entropy Search Framework for Multi-Objective Bayesian Optimization. *Journal of Artificial Intelligence Research (JAIR)*, 72: 667-715, 2021.
2. [TECS'20] Nitthilan K. Jayakodi, Syrine Belakaria, Aryan Deshwal, and Janardhan Rao Doppa. Design and Optimization Framework to Trade-off Energy and Accuracy of Inference on Mobile Platforms via Pretrained Neural Networks. *ACM Trans. on Embedded Computing Systems (TECS)*, 19(1): 4:1-4:24, 2020.
1. [TECS'19] Aryan Deshwal, Nitthilan K. Jayakodi, Biresh Joardar, Janardhan Rao Doppa, Partha Pratim Pande. MOOS: A Multi-Objective Design Space Exploration and Optimization Framework for NoC enabled Manycore Systems. *ACM Transactions on Embedded Computing Systems (TECS)*, 18(5s): 77:1-77:23, 2019.

Workshop Papers

7. Aryan Deshwal, Sait Cakmak, Yuhou Xia, David Eriksson. Sample-Efficient Bayesian Optimization with Transfer Learning for Heterogeneous Search Spaces. *AutoML Conference 2024 (Workshop Track)*.
6. Aryan Deshwal, Syrine Belakaria, Janardhan Rao Doppa. Scalable Combinatorial Bayesian Optimization with Tractable Statistical models. *Proceedings of NeurIPS Workshop on Engineering Modeling, Simulation, and Design, 2020*.
5. Syrine Belakaria, Aryan Deshwal, Janardhan Rao Doppa. Information-Theoretic Multi-Objective Bayesian Optimization with Continuous Approximations. *Proceedings of NeurIPS Workshop on Engineering Modeling, Simulation, and Design, 2020*.
4. Syrine Belakaria, Aryan Deshwal, Janardhan Rao Doppa. Max-value Entropy Search for Multi-Objective Bayesian Optimization with Constraints. *Proceedings of NeurIPS Workshop on Machine Learning and the Physical Sciences, 2020*.
3. Aryan Deshwal, Syrine Belakaria, Janardhan Rao Doppa, and Alan Fern. Optimizing Discrete Spaces via Expensive Evaluations: A Learning to Search Framework. *ICML Workshop on Real World Experiment Design and Active Learning, 2020*.
2. Syrine Belakaria, Aryan Deshwal, and Janardhan Rao Doppa. Max-value Entropy Search for Multi-Objective Bayesian Optimization. *ICML Workshop on Real World Experiment Design and Active Learning, 2020*.
1. Syrine Belakaria, Aryan Deshwal, and Janardhan Rao Doppa. Uncertainty aware Search framework for Multi-Objective Bayesian Optimization with Constraints. *ICML Workshop on Automated Machine Learning (AutoML), 2020*.

CONFERENCE/INVITED TALKS

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- [How AI is Accelerating Scientific Discovery @ Upper Bound, 2026](#).
 - Data-driven approaches for physical models workshop in School of Mathematics, University of Minnesota, 2025
 - AI to Accelerate Scientific Discovery and Engineering Design @ Carleton College, 2025
 - Bayesian Optimization over Combinatorial Inputs @ Meta's Adaptive Experimentation Workshop, 2025
 - Adaptive Experimental Design to Accelerate Scientific Discovery and Engineering Design, MnRI seminar, UMN 2025
 - Adaptive Experimental Design To Accelerate Scientific Discovery and Engineering Design @ Oregon State University, 2024.
 - Adaptive Experimental Design to Accelerate Scientific Discovery and Engineering Design @ Indian Institute of Science, 2024.

- Adaptive Experimental Design to Accelerate Scientific Discovery and Engineering Design @ Indian Institute of Technology, Roorkee, 2024.
- Adaptive Experimental Design to Accelerate Scientific Discovery and Engineering Design @ Indian Institute of Technology, Delhi, 2024.
- Invited Talk, Bayesian Optimization over Combinatorial Structures @ INFORMS Conference, 2023.
- Invited Talk, Bayesian Optimization over Combinatorial Structures @ Amazon Science Seminar, 2023.
- Bayesian Optimization over High-Dimensional Combinatorial Spaces via Dictionary-based Randomized Continuous Embeddings @ AISTATS, 2023.
- Invited Talk, [Bayesian Optimization over Combinatorial Structures @ Secondmind](#), 2022.
- Invited Talk, [Bayesian Optimization over Combinatorial Structures @ Valence Labs](#), 2022.
- Invited Talk, Bayesian Optimization over Combinatorial Structures, Invited Talk @ Argonne National Laboratory, 2022.
- Bayesian Optimization over Permutation Spaces @ AAAI, 2022.
- Bayesian Optimization over Hybrid Spaces @ ICML, 2021.
- Mercer Features for Efficient Combinatorial Bayesian Optimization @ AAAI, 2021.
- Optimizing Discrete Spaces via Expensive Evaluations: A Learning to Search Framework @ AAAI, 2020.
- MOOS: A Multi-Objective Design Space Exploration and Optimization Framework for NoC enabled Many-core Systems @ Embedded Systems Week Conference, 2019.
- Randomized Greedy Search for Structured Prediction: Amortized Inference and Learning @ IJCAI, 2019.
- Learning and Inference for Structured Prediction: A Unifying Perspective @ IJCAI, 2019.

TEACHING EXPERIENCE

Tutorial Instructor

- Lead organizer of Half-day Tutorial on Black Box Optimization from Offline Datasets upcoming at AAAI 2026. [Slides](#).
- Lead organizer of *AAAI Tutorial on Recent Advances in Bayesian Optimization* at AAAI-2023. [Slides](#) [Link](#).
- Lead organizer of *IJCAI Tutorial on Bayesian Optimization: From Foundations to Advanced Topics* at IJCAI-2022.
- Lead organizer of *AAAI Tutorial on Bayesian Optimization: From Foundations to Advanced Topics* at AAAI-2022.

Teaching at UMN

- CSCI 5525: Advanced Machine Learning, Fall 2025, Fall 2026
- CSCI 5980/8980: AI for Sequential Decision Making, Spring 2025, Spring 2026

INDUSTRIAL RESEARCH EXPERIENCE

Research Intern <i>Meta Research - Adaptive Experimentation Team</i>	June 2023–August 2023 <i>San Francisco</i>
Research Intern <i>Meta Research - Adaptive Experimentation Team</i>	June 2022–August 2022 <i>New York</i>
Research Intern <i>Google Research - Vizier Team</i>	May 2021–August 2021 <i>Remote</i>
Research and Development Engineer <i>Works Applications</i>	October 2017–May 2018 <i>Tokyo, Japan</i>
Software Engineering Intern <i>Samsung</i>	June 2016–August 2016 <i>Bengaluru, India</i>

CONFERENCE/WORKSHOP ORGANIZATION

- Co-organizer of the [Decision-Making from Offline Datasets to Online Adaptation: Black-Box Optimization to Reinforcement Learning Workshop](#) at ICML, 2026.
- Lead organizer of *5th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI for Biological Sciences Theme)* at AAAI-2026. [Website Link](#).
- Lead organizer of *4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI for Biological Sciences Theme)* upcoming at AAAI-2025. [Website Link](#).
- Lead organizer of *3rd Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI for Materials and Manufacturing Theme)* at AAAI-2024. [Website Link](#).
- Lead organizer of *2nd Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI for Earth and Environment Science Theme)* at AAAI-2023. [Website Link](#).
- Lead organizer of *1st Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI for Chemistry Theme)* at AAAI-2022. [Website Link](#).

PROFESSIONAL SERVICE AND OUTREACH ACTIVITIES

Area Chair and Senior Program Committee Member

- Forty-First AAAI Conference on Artificial Intelligence (AAAI), 2027
- Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026
- Forty-Third International Conference on Machine Learning (ICML), 2026
- Fourteenth International Conference on Learning Representations (ICLR), 2026
- Twenty-Ninth International Conference on Artificial Intelligence and Statistics (AISTATS), 2026
- Fortieth AAAI Conference on Artificial Intelligence (AAAI), 2026
- Forty-second International Conference on Machine Learning (ICML), 2025
- Twenty-Eighth International Conference on Artificial Intelligence and Statistics (AISTATS), 2025
- Thirteenth International Conference on Learning Representations (ICLR), 2025
- Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI), 2025
- Thirty-Eighth AAAI Conference on Artificial Intelligence (AAAI), 2024
- Thirty-Seventh AAAI Conference on Artificial Intelligence (AAAI), 2023

Program Committee Member

- Thirty-ninth Conference on Neural Information Processing Systems (NeurIPS), 2025 (Top Reviewer award)
- Thirty-eighth Conference on Neural Information Processing Systems (NeurIPS), 2024 (Top Reviewer award)
- Forty-first International Conference on Machine Learning (ICML), 2024
- Twelfth International Conference on Learning Representations (ICLR), 2024
- Thirty-Eighth AAAI Conference on Artificial Intelligence (AAAI), AI for Social Impact Track, 2024
- Thirty-Seventh Conference on Neural Information Processing Systems (NeurIPS), 2023
- Fortieth International Conference on Machine Learning (ICML), 2023
- International Conference on Learning Representations (ICLR), 2023
- Twenty-Sixth International Conference on Artificial Intelligence and Statistics (AISTATS), 2023
- Thirty-Sixth Conference on Neural Information Processing Systems (NeurIPS), 2022
- Thirty-Ninth International Conference on Machine Learning (ICML), 2022
- Thirty-Fifth Conference on Neural Information Processing Systems (NeurIPS), 2021

- Thirty-Eighth International Conference on Machine Learning (ICML), 2021 [Top Reviewer Award](#), [Expert Reviewer](#)
- Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI), 2021
- International Conference on Learning Representations (ICLR), 2021 [Outstanding Reviewer Award](#)
- Thirty-Fourth Conference on Neural Information Processing Systems (NeurIPS), 2020
- Thirty-Seventh International Conference on Machine Learning (ICML), 2020 [Top Reviewer Award](#)
- Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI), 2020

Conference and Journal Reviewer

- Reviewer for NeurIPS-2019, UAI-2019, KDD-2019, AISTATS-2019, ESWEK-2019, AAAI-2020, AISTATS-2020, UAI-2020, KDD-2020, UAI-2021, AISTATS-2022, NeurIPS-2022, ICLR-2023, AISTATS-2023
- Transactions on Machine Learning Research (TMLR), 2022
- Journal of Artificial Intelligence Research (JAIR), 2019, 2021

OPEN-SOURCE SOFTWARE

- [GAUCHE](#): Gaussian process library for chemistry.
- Bayesian Optimization Software: [PSR](#), [MerCBO](#), [HyBO](#), [BOPS](#), [BODi](#), [BO_of_COFS](#), [LADDER](#), and [Heterogeneous.botl](#).